Airline schedule planning: a review and future directions

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Received 6 September 2016
Revised 26 October 2016
Accepted 27 October 2016

Abstract

Purpose – The purpose of this paper is twofold: first to carry out a comprehensive literature review for state of the art regarding airline schedule planning and second to identify some new research directions that might help academic researchers and practitioners.

Design/methodology/approach – The authors mainly focus on the research work appeared in the last three decades. The search process was conducted in database searches using four keywords: "Flight scheduling," "Fleet assignment," "Aircraft maintenance routing" (AMR), and "Crew scheduling". Moreover, the combination of the keywords was used to find the integrated models. Any duplications due to database variety and the articles that were written in non-English language were discarded.

Findings – The authors studied 106 research papers and categorized them into five categories. In addition, according to the model features, subcategories were further identified. Moreover, after discussing up-to-date research work, the authors suggested some future directions in order to contribute to the existing literature.

Research limitations/implications – The presented categories and subcategories were based on the model characteristics rather than the model formulation and solution methodology that are commonly used in the literature. One advantage of this classification is that it might help scholars to deeply understand the main variation between the models. On the other hand, identifying future research opportunities should help academic researchers and practitioners to develop new models and improve the performance of the existing models.

Practical implications – This study proposed some considerations in order to enhance the efficiency of the schedule planning process practically, for example, using the dynamic Stackelberg game strategy for market competition in flight scheduling, considering re-fleeting mechanism under heterogeneous fleet for fleet assignment, and considering the stochastic departure and arrival times for AMR.

Originality/value – In the literature, all the review papers focused only on one category of the five categories. Then, this category was classified according to the model formulation and solution methodology. However, in this work, the authors attempted to propose a comprehensive review for all categories for the first time and develop new classifications for each category. The proposed classifications are hence novel and significant.

Keywords Aircraft maintenance routing, Crew scheduling, Fleet assignment, Flight scheduling

Paper type Research paper

1. Introduction

The airline industry is characterized by tight competitive market, high operational cost, variable passenger demand, heavy traffic, and strong regulations. In this situation, airline companies have to efficiently manage their resources that include flights, aircraft, and crews (Sherali *et al.*, 2013a). In order to manage these resources, airline schedule planning problems are solved by the airline companies, while considering the vast number of regulations related to aircraft and crews that result in complex and nontractable problems. Consequently, airline scheduling is decomposed into four stages: the flight scheduling problem (FSP), the fleet assignment problem (FAP), the aircraft maintenance routing problem (AMRP), and finally the crew scheduling problem (CSP). Traditionally, these problems are solved sequentially, where the solution in each stage is used as an input for the subsequent stage, as shown in Figure 1.

Figure 1 describes the sequential operations that take place before the departure of an aircraft. At the beginning, the flight schedule is constructed considering marketing issues such as passenger demand and ticket price (Yan *et al.*, 2007; Lee *et al.*, 2007). Thereafter, each flight is covered by the specific aircraft type, and feasible maintenance routes are constructed

Industrial Management & Data Systems
Vol. 117 No. 6, 2017
pp. 1201-1243

0263-5577
DOI 10.1108/IMDS-09-2016-0358

for each fleet. These two steps are performed by fleet assignment (Rexing *et al.*, 2000; Barnhart *et al.*, 2002; Barnhart *et al.*, 2009) and aircraft maintenance routing (AMR) (Gopalan and Talluri, 1998; Sriram and Haghani, 2003; Sarac *et al.*, 2006; Başdere and Bilge, 2014). In the last stage, cabin crew and cockpit are assigned to each flight in order to form an anonymous pairing, while satisfying the regulations and contractual issues (Vance *et al.*, 1997; Ehrgott and Ryan, 2002; Zeghal and Minoux, 2006; Muter *et al.*, 2013). The generated pairings are grouped in order to form personal rosters, considering vacations and crew requests. Although there are interrelations between each stage, they are usually solved sequentially due to their complexity. The sequential approach leads to suboptimality solution, which means the solution is optimal in one stage and not in others. In order to avoid this problem, scholars now pay much attention to solving more integrated airline scheduling models so as to ameliorate the solution quality and the anticipated profit of the airline companies (Mercier *et al.*, 2005; Haouari *et al.*, 2009; Sherali *et al.*, 2013b; Cacchiani and Salazar-González, 2016). The airline schedule planning process is one of the challenging processes faced by airline companies since the vast number of regulations and rules need to be considered for each resource in the process. For aircraft, the maintenance requirements and number of flying hours should be satisfied during the planning. For crew members, union rules, restrictions on the flying time, and contractual issues should be respected in the planning process.

In the literature, it is observed that there is no effort can be found to cover all the stages of airline schedule planning. Most of contributions focus on one stage only, as shown by Etschmaier and Mathaisel (1985), Sherali *et al.* (2006), and Gopalakrishnan and Johnson (2005) who focused on flight scheduling, fleet assignment, and crew scheduling, respectively. In these studies, the focused stage was classified according to the model formulation and solution methodology, which make it hard to relate and understand the main operational features of each research work. As a result, there is general lack of comprehensive view that considers all the stages of airline schedule planning and discusses their features and operational considerations. Therefore, in this paper, we focus on the nature of the problem and its operational considerations and classify each stage based on the problem characteristics rather than solution methodology and formulation methods. The main benefit of this classification over existing classification lies in its ability to give precise and in-depth sub-categories of each problem in airline schedule planning, thereby providing an insight into the progress of this field. So, the major contribution of this paper is to fill this gap and to facilitate the future work in this topic through two steps: first by providing a map of what has been done in airline schedule planning by presenting an overview of the current state of the art regarding airline schedule planning processes and second by identifying some fertile opportunities for future research for researchers who are interested in the airline schedule planning field.

The remaining of this paper is organized as follows. In Section 2, the research method is described. In Section 3, the flight scheduling is presented. Section 4 presents the FAP. In Section 5, we describe the AMRP. The CSP and its corresponding problems are discussed in Section 6. In Section 7, we survey the integrated airline schedule planning models. Finally, some concluding remarks and future research directions are presented in Section 8.

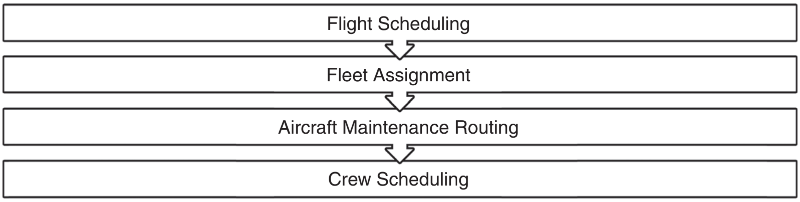

Figure 1.
The airline schedule planning processes

2. Research method

2.1 Source of literature

The main objective of this research work is to provide an overview of the current state of the art regarding airline scheduling planning processes and identify some new research directions. In order to achieve this goal, we searched using three main sources: ScienceDirect – Online Journals by Elsevier Science, Emerald, and Springer LINK Online Libraries. This survey is conducted based on articles in top journals, limited number of conference proceeding, handbooks, and theses.

As we mentioned that we need to provide a comprehensive survey for airline schedule planning, we mainly focused on the papers published in the last three decades. Moreover, any paper that cited significant contribution and out of this time frame is added to our survey, in order to enrich our review paper.

We conducted the search process using four keywords: "Flight scheduling," "Fleet assignment," "AMR," and "Crew scheduling" in all database listed above. Moreover, we searched by using combination of keywords in order to find the integrated airline schedule planning models. During our search process, the papers were screened based on their quality, which was determined according to the impact factor of the journal, the total number of citations, and the journal ranking. In fact, our screening process was limited to the journals ranked as Q1, Q2, and Q3. In particular, we read each of the screened papers and added any new article that did not appear in our first search. This allowed us to collect up-to-date research in our field. Any duplications due to database overlap and non-English language papers were discarded. Also, any paper that matched with the keywords but whose scope was not relevant to our categories was rejected.

The approach adopted in this paper is similar to the ones used by Martínez-Costa *et al.* (2014) and Chung *et al.* (2015) who provided a review for strategic capacity planning in manufacturing and disruption management in airline operations, respectively.

After making that search, 116 articles were left, among which, there were 113 journal papers, one conference proceeding, one book chapter, and one thesis. In fact, the journal papers came from 27 different journals, even though 75 percent of them came from top journals, as shown in Table I. Table I illustrates the journal used in the literature survey, the number of papers picked from each journal, the impact factor of each journal, and the number of citation. The content of this table indicates how good the papers are, as most of the papers came from top journals in the airline transportation planning.

2.2 The philosophy of the review work

The review work was conducted by following these four steps:

(1) identify the relevant literature concerning the airline schedule planning, and identify how different classes of problems varied in terms of operational considerations;

Journal	Number of papers	Impact factor	Total citation	Journal ranking
Transportation Science	34	3.295	3,945	Q1
European Journal of Operational Research	18	2.679	31,744	Q1
Operations Research	10	1.777	10,200	Q2
Computers & Operations Research	10	1.988	7,545	Q1
Management Science	4	2.741	22,776	Q1
Transportation Research Part B	4	3.769	7,358	Q1
Transportation Research Part A	3	1.994	5,024	Q2
Computers & Industrial Engineering	2	2.086	6,357	Q2
Transportation Research Part C	1	3.075	4,153	Q1

Note: The data collected in this table delivered from ISI web of knowledge

Table I. The features of the journals used in the literature survey

(2) categorize the problem into five main group: flight scheduling, fleet assignment, AMR, crew scheduling, and the integrated models;

(3) define subgroup for each main group according to the predetermined variants, discuss the main features of each subgroup, and summarize the main characteristics of each subgroup's models (objective function, solution methods, and used data); and

(4) suggest the new wave of research in the airline schedule planning.

2.3 Classification schemes

There are many classification schemes in the literature to categorize the airline schedule planning. Using different solution methodology (e.g. exact methods and meta-heuristics), and formulation methods (e.g. set-partitioning, multi-commodity network flow, and integer programming) are the most common used classification schemes in the previous efforts to produce an airline schedule planning classification scheme. Since our focus on the nature of the problem and its operational considerations, our classification scheme is based on the problem characteristics rather than solution methodology and formulation methods. The main benefit of this classification scheme over existing scheme lies in its ability to give precise and in-depth subcategories of each problem in airline schedule planning, thereby providing an insight into the progress of this field.

3. FSP

The first category in this paper is FSP, which is considered to be the first problem to be solved by airline companies before operation commences. FSP aims to generate a timetable that contains a list of flights with their corresponding origin, destination, departure time, and arrival time, while considering some marketing issues such as passenger demand and ticket price (Yan *et al.*, 2007; Lee *et al.*, 2007). Usually, the flight schedule is constructed at least six months in advance of the scheduled flight, with the objective of maximizing the expected profit. The FSP is usually solved in two steps; in the first step, the timetable is constructed, while evaluation of the generated timetable is carried out in the second step. The timetable evaluation is made by checking the feasibility of each flight leg and ensuring cost minimization of each flight. In order to improve the constructed timetable, these two steps are repeated until no more improvements can be made. In order to build an efficient flight schedule timetable, the airline companies have to determine certain aspects such as the projected markets served, the type of flights (non-stop flights or connecting flights), and the frequency of the flights in each market.

After reading FSP papers, the flight scheduling models can be classified into two main subcategories. The first subcategory includes the models that consider market share and passenger demand fluctuations, as described in Section 3.1. The second subcategory groups the models that pay attention to the robustness of the generated timetable, as described in Section 3.2. In Section 3.3, Table II provides a summary of the models' characteristics in chronological order. In Section 3.4, we provide a discussion for the reviewed FSP papers.

3.1 Passenger demand and market share in FSP

To build an efficient flight schedule, passenger demand fluctuations and the market share should be considered in the planning stage. Yan and Young (1996) were among the first authors who considered the expectation of demand variation, which enables the planning managers to modify the flight schedule that is built based on fixed passenger demand. The flight timetable adjustment can be done in different ways: deleting of multi-stop flight legs, or changing the departure time, or renting aircraft. The proposed model was tested by a case study on the international operations of a major Taiwan airline, and the computational

Table II. Characteristics of flight scheduling model

Author/s (year)	Planning horizon	Network	Model formulation	Objective function	Solution procedure	Computational study: Data	Computational study: Airline	Other features of the model: Stochastic	Robustness	Passenger demand: Fixed	Passenger demand: variable	Market share: Fixed	Market share: Variable
Yan and Young (1996)	W	TSN	MCNF	Min scheduling and fleeting costs	Lagrangian relaxation and Lagrangian heuristic	RL	Taiwan			✓			
Yan and Tseng (2002)	NA	TSN	MCNF	Min passenger and fleeting costs	Developed algorithm	RL	Taiwan			✓		✓	
Yan *et al.* (2007)	D	TSN	NLMIP	Min passenger and fleeting costs	Developed heuristic	RL	Taiwan	✓		✓			✓
Lee *et al.* (2007)	W	–	MILP	Min percentage of late arrivals and flight time credit.	Genetic algorithm	G	–		✓	✓			
Yan *et al.* (2008)	D	TSN	NLIP	Min fleet flows and expected costs.	Arc-based heuristic algorithm and Route-based heuristic algorithms	RL	Taiwan	✓			✓		✓
Jiang and Barnhart (2009)	D&W	TSN	MILP	Max revenue minus operating cost	ILOG CPLEX 9.0	RL	US	✓			✓		
Burke *et al.* (2010)	W	TSN	MCNF	Max Schedule flexibility and reliability	Multi-meme memetic algorithm	RL	KLM Royal Dutch	✓	✓	✓			
Sohoni *et al.* (2011)	NA	–	IP	Max The expected profit	New cut generation algorithm	RL	US carrier	✓	✓	✓			
Jiang and Barnhart (2013)	D	TSN	MILP	Max the total revenue	A decomposition-based approach	RL	US		✓				
Kepir *et al.* (2016)	D	TSN	MILP	Min the number of idle planes	Heuristic Algorithm	RL	Turkish			✓			

Notes: Planning horizon: daily (D) or weekly (W). Network: time-space network (TSN). Model formulation: NLMIP: non-linear mixed integer programming, NLIP: non-linear integer programming, MILP: mixed integer linear programming, MCNF: multi-commodity network flow, IP: integer programming. Used Data: real life (RL) or generated (G) results showed that the model outperformed the trial-and-error method used by Taiwan airlines. However, this paper has two drawbacks: the first one is the focus on one-stop and non-stop flights and the second one is that the model neglects the market share consideration. Yan and Tseng (2002) improved the paper by Yan and Young (1996) by considering the market share in their model. The proposed model was solved with the objective of maximizing the system profit, but the passenger demand and projected market share were still fixed.

In order to relax the fixed market share assumption that was set by Yan and Tseng's model, Yan *et al.* (2007) incorporated passenger demand choice to reflect a variable market share in their proposed model in order to improve the flight schedule in a competitive market. The authors developed a heuristic approach to solve the problem iteratively and showed good performance during validation testing. In this model, we can say that the authors considered the variable market share, but the passenger demand was assumed to be fixed. The paper presented by Jiang and Barnhart (2009) considered the variability of passenger demand but neglected the market share. The authors developed a dynamic scheduling model that used a flight retiming mechanism in order to re-optimize the flight and matched the stochastic passenger demand and aircraft seats. Based on data provided by a US airline, the authors reported that the annual saving increased from $18 million to $36 million.

In a trial that considered the variability of the market share and passenger demand simultaneously, Yan *et al.* (2008) developed a two-stage stochastic programming model that considered the two aspects, resulting in a highly complex model (NP-hard). In order to avoid high complexity, a single fleet with non-stop flight operation was considered in the constructed model. Based on the data provided by Taiwan airlines, comprised of eight cities covered by 19 aircraft, the computational results showed good performance.

3.2 Robust flight scheduling models

Robustness is considered one of the interesting research directions in the airline schedule planning arena. The main goal of a robust flight schedule is to make the generated timetable less sensitive to disruptions. It is a pro-active way to absorb any changes that may happen in the scheduled timetable (Chung *et al.*, 2015; Hing Kai and Alain Yee Loong, 2015). Actually, there are few research studies that pay attention to the robustness of FSP models. For example, Lee *et al.* (2007) developed a model that improved solution robustness by increasing the adjustability of the flight departure time in the generated schedule timetable. The proposed model enabled the planning mangers to avoid disruptions by retiming the flight departure time without inserting or deleting any flights. The model was evaluated by using simulation model SIMAIR2.0, and it provided a schedule with lower costs and better on-time performance, but in longer computational time. To our best knowledge, the robustness can be achieved in two different ways: the first way is by constructing a reliable model with adjustable departure time that allows using the flight retiming option as mentioned in Lee *et al.* (2007). The second way is by constructing a flexible model that offers many swapping options for the aircraft. Both options are considered by Burke *et al.* (2010) who improved the robustness of the original schedule by using the flight retiming option and the swapping option. The model's flexibility and reliability were investigated by using real data from KLM Royal Dutch Airlines, and the results showed significant improvement in both objectives.

Another recent work on robust FSP was reported by Sohoni *et al.* (2011). The authors considered the block time uncertainty as a measure of robustness. Stochastic integer programming was developed in order to optimally adjust the generated schedule to maximize the expected profit. Based on real data, the model was solved by an efficient algorithm based on cut generation techniques, and the computational results showed good performance. Recently, Jiang and Barnhart (2013) proposed a robust model, where the robustness is measured by the number potentially connected itinerary.

3.3 Characteristics of the models

This section presents other characteristics of the reviewed flight scheduling models. As the result of the survey carried out, Table II classifies the models according to the following factors: planning horizon, network representation, model formulation, objective function of the model, and solution procedure, with the authors demonstrating the computational study with real data or using generated data. Other features are presented in the last six columns of Table II.

3.4 Discussion

After reviewing the papers regarding market share and passenger demand, we can see that there is a sequence of effort that its complexity increased gradually. This sequence starts with Yan and Young (1996) and Yan and Tseng (2002) who studied the fixed market share and passenger demand. Implementing these models in reality is quite uncertain, since they fail to consider the variability of market share and passenger demand, which AR one the main characteristics of aviation industry. Later, another two studies emerged by Yan *et al.* (2007) and Jiang and Barnhart (2009) who investigated the variability of one aspect of market share and passenger demand. Their research works are promising, and they moved toward the real aspects of airline industry while considering the variability of market share or passenger demand. Finally, the peak of the efforts appeared in Yan *et al.* (2008) who presented stochastic model. The strength of this study appears while considering stochastic market share and passenger demand, which means capturing the reality. On the other hand, during the implementation, they used test cases which are very small and do not reflect the reality. Therefore, their computational experiments are not enough to show the applicability in real airline industry.

In terms of robust FSP models, few research attempts are observed by Lee *et al.* (2007), Burke *et al.* (2010), and Sohoni *et al.* (2011). These research trials are easy to be implemented since they produce a timetable that is less sensitive to disruptions.

In overall, we notice that the number FSP research work is relatively low compared to other categories, and this area still needs another research effort to be exerted to fill some research gap, as explained in the conclusion section.

4. FAP

FAP is considered the second category in our study, which plays an important role to determine which aircraft types will be selected to cover the scheduled flight legs. Moreover, the FAP has to satisfy that each flight leg should be covered by exactly one aircraft type, and the total number of aircrafts that belong to the same type does not exceed the number available. This problem is solved with two different objective functions, which maximize the profit and minimize the assignment cost. With respect to the cost, there are different types of costs included in the assignment costs, such as operating cost, carrying cost, and spill cost. Finding the balance between aircraft capacities and passenger demand is a great challenge that faces planning mangers because of great impact of fleet assignment on the airline's profit. Imagine, for example, the assignment of a large aircraft to cover a flight leg with a limited number of passengers, and aircraft with a small number of seats is assigned to cover a trip with large number of passengers. In the first case, there are spoiled seats, whereas in the second case there are spilled passengers. Both cases are not favorable for airlines, so that the FAP should be solved based on accurate passenger demand forecasts. Using such models resulted in an increase of $30 million in annual revenue of US Airways (Barnhart *et al.*, 2002).

In the following sections, we categorize the fleet assignment models (FAM) that are presented in the literature into subcategories according to the model characteristics. Basic FAM is presented in Section 4.1. In order to enhance basic FAM, researchers consider the variable departure time, network effect, and robustness, as shown in Sections 4.2, 4.3, and 4.4, respectively. In Section 4.5, we discuss the demand-driven re-fleeting FAM, whereas the weekly FAM is addressed in Section 4.6. In Section 4.7, Table II presents other characteristics of FAM. In Section 4.8, we provide a discussion for FAP research work.

4.1 Basic FAM

In this section, we present the FAMs, which constitute the base of the work in this field. These models assume that the flight schedule is solved and the timetable is generated. Also, these models are solved under the assumption that the schedule is repeated every day of the week (Daily planning horizon) in order to avoid high computational flexibility.

Abara (1989) was one of the first authors to present basic FAM based on a connection network structure (in which the nodes represent the flight leg and the arcs represent the possible connections), in order to find feasible connections or turns between two flight legs. Actually, Abara's formulation has major disadvantages, which are as follows: a large number of connections increase the model's complexity and the model fails to solve the problem for heterogeneous fleets. Implementing the model on American airlines resulted in improvement in the profitability of the company. Abara's work was extended by Rushmeier and Kontogiorgis (1997). They proposed a model that was solved by using some preprocessing methods instead of finding feasible turns or connections. The authors also considered the resource availability that is expressed by linear penalties for any violation to the preplanned resource utilization. They reported profit improvement of about \$15 million in US Airways.

Another basic FAM was presented by Hane *et al.* (1995), which was different from the underlying network structure. The presented model was formulated based on the time-space network, in which the nodes represent the departure and arrival events, whereas the arcs represent the flight legs. They proposed some preprocessing stages in order to reduce the size of the structured network, such as node aggregation, island isolation, and the removing of unnecessary ground arcs. This model can find fast solutions for real problems comprising 2,500 flight legs and 11 fleet types.

Although basic fleet FAM determines which aircraft types will be selected to cover the scheduled flight legs, it has some downsides. For example, it fails to consider the network effect and the recapture of the spilled passenger. Also, basic FAM does not consider the variability of departure times that provides flexibility to the model, and many other features that are discussed in the following sections.

4.2 FAM with variable departure time

Variable departure time provides flexibility to the FAM as shown by Levin (1971). He was the first author to consider variable departure times in a form of integer programming with binary constraints. The proposed model was solved with a branch-and-bound approach instead of dynamic programming in order to reach the optimal solution. This model does not allow for the considering of aircraft capacity and multiple fleet types. The paper by Desaulniers, Desrosiers, Dumas, Solomon, and Soumis (1997) was basically an extension of Levin's work. The authors considered heterogeneous fleet as well variable departure times in their proposed model. The authors presented two different formulations for the fleet problem, which are set-partitioning formulation, and multi-commodity network flow model formulation, and proposed different branching strategies for solving the problem. They reported profit and tractability improvement after implementing their approach. Similar to Levin's model, Rexing *et al.* (2000) proposed a model at which the departure time of each flight leg is discretized within specified time windows in order to increase the flight connection opportunities. The proposed model was solved by using two approaches: the first one was a direct approach by using commercial software which was good for speed and small-sized problem, while the second one was an iterative approach which was appropriate for large-scale problems. Actually, the authors reported potential savings of about $67,000 in the operating cost beside the spill cost in US airlines.

4.3 FAM with network effect

The network effect means that the demand that is coming from multi-leg passengers is dependent on the availability of seats on the flight legs. To our best knowledge, the PhD dissertation by Farkas (1995) was one of the first research works that attempted to address the network effect in his proposed model. Farkas's model fails to consider the recapture of a spilled passenger, which was further considered by Barnhart *et al.* (2002) who proposed an itinerary-based model. The authors used a network of 2,044 legs and 76,641 paths in order to validate their model. They reported $30 million as an increase in the annual revenue of US airlines.

The next two papers consider stochastic passenger demand which was neglected in the previous two papers. Jacobs *et al.* (2008) proposed a decomposition strategy to incorporate the network effect in the origin-destination FAM. Also, this model considered the stochastic passenger demand (demand uncertainty) by using the demand-driven dispatch technique. This model shows a 2.8 percent profit improvement over the leg FAM. The authors reported that this model was adopted by many USA, European, and Asian Airlines due to its ability to reduce the cost. Another recent work that considered stochastic passenger demand was reported by Dumas *et al.* (2009). The authors enhanced their model by using the information from the passenger flow model. The proposed model considered the network effect, the recapture between itineraries, and the stochastic demand. The authors reported profit improvement in Air Canada.

4.4 Robust FAM

The airline industry is most likely faced by disruptions and unforeseen circumstances. Hence, robust models are imperative so as to provide a solution that better withstands during disruption. Actually, robust FAM has not received much attention. Rosenberger *et al.* (2004) developed a robust model that used the concept of isolate hubs, which generates many short cycles that are less sensitive to flight cancelation. This method of formulation showed better performance over traditional FAM. Another robust FAM model was presented by Smith and Johnson (2006) who increased the model flexibility by incorporating the station purity concept, which means limiting the number of fleets or crew compatible families that can serve each station. The model addressed the crew and maintenance issues. Because of the negative computational impact of addressing the station purity, the authors used the station decomposition based on column generation to solve the model. The authors estimated an annual saving of about $100 million because of reduced maintenance and crew operational cost in a major US domestic airline.

4.5 Demand-driven re-fleeting FAM

Demand-driven re-fleeting is an approach that is used by airlines in response to demand uncertainty, operational disruptions, maintenance, and crew changes. This approach changes the initial fleet assignment based on updated demand information. As much of the forecasted data are accurate, the improvement will be significant, because there are many stages that depend on the fleet assignment, such as routing and crew assignment. Hence, it is very important to estimate demand accurately.

In this section, we review the models that address the demand-driven re-fleeting concept. Berge and Hopperstad (1993) were among the first researchers who presented the demand re-fleeting concept to capture the demand fluctuation during fleet assignment. In order to solve the re-fleeting problem in a reasonable computational time, the authors proposed two heuristics. The first heuristic was called the Sequential Minimum Cost Flow Method that can find the solution in 4 min, whereas the second heuristic was called the Delta Profit Method (DELPRO), and it reaches the solution in 3 min; however, it can find the swap opportunity in a restricted way. Based on real data provided by a US domestic carrier, the computational results showed 1-5 percent improvement in the operating profits as a consequence of applying the demand-driven re-fleeting approach. Berge and Hopperstad's work was extended by Talluri (1996) who improved the DELPRO heuristic in terms of swapping opportunities and computational time. On a data set provided by USAir, Talluri's algorithm could find ten swap opportunities in three to four seconds.

Another re-fleeting model was presented by Jarrah *et al.* (2000). The authors developed an efficient FAM that gives planning managers the opportunity to generate different feasible scenarios and select the most appropriate one. This kind of formulation avoids the manual modification made in the basic FAM during the re-fleeting process. The authors reported that the model can reach high-quality solutions in 5 min. On an attempt to reduce the representation of demand-driven re-fleeting model, Sherali *et al.* (2005) studied the polyhedral structure of the re-fleeting model in order to make dynamic reassignment, under the condition of availability of the improved demand forecasts. The model could solve moderate-sized problems with 5 percent optimal tolerance. On a real network of 5,098 flight legs and 64,247 paths, operated by United Airlines, the solution was generated within 20 min.

4.6 Weekly FAM

All the previous FAM models, except Dumas *et al.* (2009) and Berge and Hopperstad (1993) assumed that the flight schedule is repeated every day of the week. From a practical point of view, airlines permit some variations on the flight schedule of each day of the week in order to cope with demand fluctuation of different flight legs (e.g. the demand at weekend is higher than other days). In light of these facts, weekly FAMs are more practical and profitable, as shown in the paper by Bélanger *et al.* (2006). The authors presented weekly FAM with aircraft homogeneity that eased the process of ground service by using the same equipment to re-supply the aircrafts. On test instances involving up to 4,400 flight legs, the results showed a significant increase in the anticipated profit of Air Canada.

Another weekly FAM was reported by Pilla *et al.* (2008) who developed an approximation method for the profit function by using the regression splines fit. In a following paper, Pilla *et al.* (2012) developed multivariate adaptive regression splines cutting planes as a solution methodology to solve the model. This method showed some improvement over the L-shaped method in the computational time, due to fast convergence. This model can capture the demand stochasticity but fails to consider the network effect and recapture of spilled passengers.

4.7 Other characteristics of FAM

As the result of our review for FAM, Table III summarizes the characteristics of the discussed models based on the planning horizon, network representation, objective function, and solution procedures. Also, the model's computational study and other features are presented in the same table.

4.8 Discussion

We observe significant efforts exerted to cover FAP in the literature since 1971. These efforts started by developing basic FAM, as shown by Abara (1989), Hane *et al.* (1995), and Rushmeier and Kontogiorgis (1997). Although these models show a good contribution to the body of knowledge, they have some practical limitations. First, they assume that the flight

Table III. Characteristics of fleet assignment models

Author/s (year)	Planning horizon	Network	Model formulation	Objective function	Solution procedure	Computational study: Data	Computational study: Airline	Other features: VDT	NE	SR	RM	DR	SD	HF
Levin (1971)	D	TSN	ILP	Min the fleet size	Branch-and-Bound methods	NA	–	✓						
Soumis *et al.* (1980)	D	TSN	MILP	Min the assignment and spill cost	Frank-Wolfe algorithm	NA	–							
Daskin and Panayotopoulos (1989)	D	TSN	ILP	Max the profit	Lagrangian relaxation approach	G	–							
Abara (1989)	D	CN	MCNF	Max revenue minus operating cost	NA	NA	–							
Balakrishnan *et al.* (1990)	D	TSN	MCNF	Max the profit	Lagrangian relaxation-based solution procedure	G	–							
Berge and Hopperstad (1993)	W	TSN	MCNF	Max the profit	1st: Sequential Minimum Cost Flow Method (SMCF) 2nd: Delta Profit Method (DELPRO)	RL	US domestic carrier					✓	✓	
Subramanian *et al.* (1994)	D	TSN	MILP	Min the operating and passenger cost	OB1 interior point code and OSL mixed integer programming code	RL	Delta Air Lines							
Hane *et al.* (1995)	D	TSN	MCNF	Min the assignment cost	Interior-point algorithm and branching	RL	–							
Talluri (1996)	D	CN	MCNF	Min the swapping cost	Developed algorithm	RL	USAir					✓		✓
Rushmeier and Kontogiorgis (1997)	D	CN	MCNF	Max the profit	Software based on Linear programming relaxation, CPLEX, and rounding heuristic	RL	USAir							

(continued)

Table III.

Author/s (year)	Planning horizon	Network	Model formulation	Objective function	Solution procedure	Computational study: Data	Computational study: Airline	Other features: VDT	NE	SR	RM	DR	SD	HF
Desaulniers, Desrosiers, Dumas, Solomon and Soumis (1997)	D	TSN	SP and MCNF	Max anticipated profit	1st: Linear relaxation and application of column generation. 2nd: Linear relaxation and application of Dantzig-Wolfe decomposition	RL	1st:North-American carrier 2nd: European carrier	✓						✓
Rexing *et al.* (2000)	D	TSN	MILP	Min the assignment cost	1st: Direct solution approach 2nd: Iterative solution approach	RL	US	✓						
Jarrah *et al.* (2000)	D	TSN	MCNF	Max the revenue minus operating cost	Developed algorithm	RL	United Airlines					✓		
Barnhart *et al.* (2002)	D	TSN	MILP	Min spill, carrying, and operating cost	Developed algorithm	RL	US		✓	✓				
Rosenberger *et al.* (2004)	NA	TSN	MCNF	Min the assignment cost	Developed algorithm	G&RL	United Airlines				✓			
Sherali *et al.* (2005)	NA	TSN	MILP	Max revenue	Polyhedral analysis-based approach	G&RL	United Airlines					✓		
Bélanger, Desaulniers, Soumis, Desrosiers and Lavigne (2006)	W	TSN	MCNF	Max profit minus fixed cost and penalties	1st: CPLEX 6.5 2nd: Developed Two-phase heuristic approach	RL	Air Canada						✓	
Bélanger, Desaulniers, Soumis and Desrosiers (2006)	D	TSN	MCNF	Max profit minus fixed cost and penalties	Branch-and-price approach	RL	North-American carrier							
Smith and Johnson (2006)	D&W	TSN	MCNF	Max the profit	1st: Column generation and primal-dual method 2nd: Fix-and-price heuristic	RL	US				✓			

(*continued*)

Table III.

Author/s (year)	Planning horizon	Network	Model formulation	Objective function	Solution procedure	Computational study: Data	Computational study: Airline	Other features: VDT	NE	SR	RM	DR	SD	HF
Jacobs *et al.* (2008)	D	TSN	MCNF	Max overall profit	Developed algorithm	RL	American Eagle		✓				✓	
Barnhart *et al.* (2009)	D	TSN	MILP	Max the expected profit contribution	Developed algorithm	RL	US							
Dumas *et al.* (2009)	W	TSN	MCNF	Min the assignment cost and revenue loss	Developed algorithm	RL	Air Canada		✓	✓			✓	
Ozdemir *et al.* (2012)	D	TSN	ILP	Min the assignment cost	Lindo 6.1	RL	Turkish			✓				
Pilla *et al.* (2012)	W	TSN	MCNF	Max the expected profit	1st: L-shaped method 2nd: Multivariate adaptive regression splines cutting plane method	RL	–						✓	
Li and Tan (2013)	D	TSN	ILP	Max the revenue	Genetic algorithm	G	–							
Wang *et al.* (2015)	D	TSN	NMIP	Max the profit	Developed algorithm	RL	–							

Notes: Planning horizon: daily (D) or weekly (W). Network: connection network (CN) or time-space network (TSN). Model formulation: multi-commodity network (MCNF), set-partitioning (SP), integer linear programming (ILP), non-linear mixed integer programming (NMIP) or mixed integer linear programming (MILP). Used Data: real life (RL) or generated (G). Other features: variable departure time (VDT), network effect (NE), spill recapture (SR), robust model (RM), dynamic re-fleeting (DR), stochastic demand (SD), Heterogeneous fleet (HF) schedule is repeated every day, which is far from reality. Second, they assume that the departure time for the flight leg is fixed and the passenger demand is deterministic. These assumptions reduce the applicability of these models in real aspects. These drawbacks of the basic FAM motivate the researchers to continue their effort to improve the models' applicability, as shown by Desaulniers, Desrosiers, Dumas, Solomon and Soumis (1997), and Rexing *et al.* (2000). These models' applicability is improved by considering the flexible departure time, which allows the mangers to make changes for the fleet assignments. However, these models have a downside, which is neglecting some operational considerations such as maintenance, crew, gate availability, and aircraft noise issues.

The improvement over basic FAM is continued by Barnhart *et al.* (2002) who considered the network effect that improves the model profitability. Their model shows significant profit improvement, but they use daily schedule which is not preferable by airline industry. Recently, the researchers paid attention to relax the deterministic assumption made by basic FAM as appeared in the works by Jacobs *et al.* (2008), Dumas *et al.* (2009), and Pilla *et al.* (2012). These models can be easily implemented since they consider the network effect beside the stochastic demand, which enable capturing the real passenger demand.

Despite the great effort that was observed while covering FAP, we notice few research works discussed the FAP robustness as a separate problem, as shown by Rosenberger *et al.* (2004) and Smith and Johnson (2006). Their models provide a promising robust plan that better withstands during the disruption, but the models fail to consider the variable departure time, which is very important for real implementation. In contrast to FAP roundness, re-fleeting models received much attention from scholars, and their models show good performance in reasonable computational time as appeared in Talluri (1996) and Jarrah *et al.* (2000).

A close look at Table III shows FAP with heterogeneous fleet received less attention from the researchers as in Talluri (1996), because it increases the model complexity. Also, the weekly FAP was discussed by few studies like Bélanger *et al.* (2006). They proposed a weekly model which is practical and profitable for airline companies.

5. AMRP

After assigning each aircraft type or fleet to the scheduled flight legs, the airline company has to assign each specific aircraft to the scheduled flight legs, which can be done by solving the AMRP. This problem is considered the third category of the review paper. In AMRP, it is assumed that the previous stages, which are FSP and FAP, are already solved. AMRP aims to determine maintenance feasible route or the sequence of flight legs, which should be flown by each aircraft (tail number). The maintenance issues play an important role in determining the route of each aircraft, because each aircraft has to undergo maintenance checks after a certain period. The maintenance checks, which are mandated by the Federal Aviation Administration (FAA), vary according to their frequencies and duration (Clarke *et al.* (1997)). There are four types of maintenance checks, which should be considered by the airline operators to avoid any violations that may result in penalty fees. The first type is a Type A check and should be performed every 65 flight-hours, involving inspection of major parts, such as an aircraft's engine. The second type is a Type B check and is performed every 300-600 flight-hours, such as visual inspection and lubrication issues. The remaining two checks are Type C and Type D; they are carried out once every one to four years, respectively. Usually, the airline operates their maintenance checks under different regulations, which are more stringent than checks imposed by FAA. For example, each aircraft has to go for the transit check maintenance every 35-40 flying hours, where the visual inspection and minimum equipment list check are carried out. Another kind of maintenance is called the balance check. AMRP needs to incorporate the Type A maintenance check, as it is the most frequent one compared to other maintenance checks.

In the following sections, we review the AMR models by discussing the different concerns, such as maintenance location and lines of flying (LOF), as shown in Sections 5.1 and 5.2, respectively. Also, different formulations are presented, i.e. string-based models are discussed in Section 5.3, network-based formulation is presented in Section 5.4, and Section 5.4 presents asymmetric traveling salesman formulations. In Section 5.6, we present the remaining time consideration. Table IV summarizes the features of the reviewed models, as shown in Section 5.7. In Section 5.8, we provide a discussion regarding the AMRP models.

5.1 Maintenance location of AMR

In this section, we discuss the models that focus on finding the maintenance stations for each aircraft in order to meet maintenance requirements (each aircraft has to undergo maintenance once every four days). Feo and Bard (1989) presented the first integrated model of routing decisions and the maintenance base location, with the objective of minimizing the maintenance cost. Their proposed model can find the optimal number and location of maintenance stations that are required for the scheduled flight legs. Their model was cast as a multi-commodity network flow problem with integer restriction on the variables, and the model was solved by a two-phase heuristics approach that reached the solution in 11 min. Sriram and Haghani (2003) extended Feo and Bard's work through considering the Type B maintenance check. An effective heuristic was developed to solve the problem in a reasonable computational time when compared with CPLEX.

5.2 LOF of AMR

The previous two models' focus was to find the maintenance stations rather than finding the route for each aircraft. In this section, we present another subcategory of papers that focus on determining the route of each aircraft by using the LOF concept. A LOF is a sequence of flight legs starting and ending at airports where the aircraft undergoes maintenance at night. In order to solve the problem by using this concept, all feasible LOFs should be generated and the model will select the optimal among them. Kabbani and Patty (1992) were among the first researchers to use the concept of LOF to solve three-day AMR. The authors formulated the AMR as a set-partitioning problem, where the rows and columns represent flights and routing, respectively. Gopalan and Talluri (1998) expanded the use of LOF to solve the k-days AMR. They developed an innovative polynomial time algorithm in order to determine maintenance feasible routes for aircrafts. The proposed algorithm was used to solve the static and dynamic formulations of the problem, where the routes were assumed to be fixed or allowed to be changed, respectively. The two-day formulation of the problem was shown to be NP-Hard when the routes were allowed to be changed. Talluri (1998) merged the three-day maintenance routing algorithm and polynomial time algorithm developed by Gopalan and Talluri (1998) in order to develop an effective heuristic to solve the four-day AMRP.

5.3 String-based models of AMR

One of the most powerful ways to find the best route for each aircraft in the fleet is using the string-based approach. A string can be defined as a sequence of connected flights that starts and ends at a maintenance station, satisfying the flow balance and maximum flying time for each aircraft. The first string-based model was developed by Barnhart *et al.* (1998) who proposed an integrated model of fleet assignment and aircraft routing. The presented model can be used to solve the AMR only, by setting the number of fleet to one. The authors formulated the AMR as a set-partitioning problem with side constraint and solved the problem by adoption a branch-and-price approach.

Table IV.
Characteristics of aircraft maintenance routing models

Author/s (year)	Planning horizon	Network	Model formulation	Objective function	Solution procedure	Computational study		Other Features Maintenance checks				Main consideration	
						Data	Airline	A	B	T	BL	OP	TL
Feo and Bard (1989)	4D	CN	MCNF	Min maintenance cost	Chvatal's set covering heuristic	RL	American	✓ (Night)				✓	
Kabbani and Patty (1992)	3D	–	SP	Min total cost	Developed heuristic	RL	American	✓					✓
Clarke *et al.* (1997)	NA	TSN	ATS	Max through value	Lagrangian relaxation and subgradient optimization	RL	US						✓
Talluri (1998)	4D	–	EL	Finding feasible routes	3 day algorithm and Polynomial time algorithm	NA	–			✓ (Night)			✓
Gopalan and Talluri (1998)	3D	CN	EL	Finding feasible routes	Polynomial time algorithm	RL	–			✓ (Night)	✓ (Night)		✓
Barnhart *et al.* (1998)	W	CN	SP	Min maintenance cost of selected strings	Branch and price, and cut algorithm	G	–						✓
Mak and Boland (2000)	NA	CN	ATS	Max utilization of Remaining flying time	Simulated annealing, Lagrangian heuristic, and subgradient optimization	G	–						✓
Sriram and Haghani (2003)	W	CN	MCNF	Min maintenance and re-assignment cost	Hybrid of random search and depth first search	G	–	✓ (Night)	✓ (Night)			✓	
Sarac *et al.* (2006)	D	CN	SP	Min number of unused legal flying hours	Branch and price approach	RL	US	✓	✓			✓	
Liang *et al.* (2011)	D	TSN	NF	Min the connection cost and maintenance cost	CPLEX callable library version 10.0	RL	US carrier			✓ (Night)			✓

(continued)

Table IV.

Author/s (year)	Planning horizon	Network	Model formulation	Objective function	Solution procedure	Computational study		Other Features Maintenance checks				Main consideration	
						Data	Airline	A	B	T	BL	OP	TL
Liang and Chaovalitwongse (2012)	W	TSN	NF	Min Total penalty cost	Diving heuristic	RL	US			✓ (Night)			✓
Haouari *et al.* (2012)	D	CN	NAFF	Finding feasible routes	CPLEX 12.1	RL	US	✓				✓	
Başdere and Bilge (2014)	W	CN	MCNF	Max utilization of Remaining flying time	1st : Branch-and- bound 2nd : Compressed annealing heuristic	RL	–					✓	

Notes: Planning horizon: daily (D), three or four days (3D, 4D) or weekly (W). Problem formulation: network flow (NF), multi-commodity network flow (MCNF), Euler tour (ET), asymmetric traveling salesman (ATS), set-partitioning (SP) or node arc flow formulation (NAFF). Maintenance checks: type A (A), type B (B), transit (T), or Balance (BL). Main consideration: tactical (TL) or operational (OL). Used Data: real life (RL) or generated (G)

Another string-based model was presented by Sarac *et al.* (2006) who proposed an operational model for AMR. They incorporated into the model some important aspects that were ignored by other models, such as resource availability constraints (human resources and maintenance stations). As in Barnhart *et al.* (1998), the problem was modeled as a set-partitioning problem and was solved by using branch-and-price method.

The major disadvantage of the string-based approach is the large number of generated strings, which require using the column generation approach at each node of the branch-and-bound tree space. This situation leads to long computational time, which is not preferred by airline companies.

5.4 Network-based models of AMR

More recently, the network-based approach was developed to avoid the drawbacks of the string-based models. This approach has many advantages, such as it has compact representation, can solve real life problems in a reasonable time, and can be tractable by commercial software.

Network-based formulation first emerged in a paper presented by Liang *et al.* (2011). They developed a new rotation-tour time-space network and incorporated new arcs for representation through value and penalties values. The new model was very compact as its size was polynomial compared to the string model; this feature makes the model easy to be solved by commercial software. However, their model has many limitations. For example, it assumes that the maintenance is performed only at night, which is far from reality. Also, their computational experiments are not large scale enough to show the scalability and applicability of the proposed model, since they only solved a network with 352 flights and 70 aircraft. In a follow-up paper, Liang and Chaovalitwongse (2012) expanded their previous work, by considering a more realistic and practical problem, the weekly AMR. The model was tested based on a real network of 5,700 flights and 550 aircraft, and the solution was reached within 5 min.

5.5 Asymmetric traveling salesman formulation of AMR

The AMR can be formulated as the asymmetric travelling salesman (ATS) problem due to similarities between the two problems, as shown in a paper by Clarke *et al.* (1997). The authors developed a model that aimed to find feasible maintenance rotations that maximize the through value (obtained from assigning one aircraft to cover certain connected flight legs). The authors fail to consider all the operational maintenance requirements. Also, Mak and Boland (2000) formulated the AMR as ATS model and solved it by adopting the simulated annealing method besides Lagrangian relaxation. The proposed approaches could reach the feasible solutions within 15-20 min.

5.6 Remaining time consideration of AMR

Remaining time is one of the important aspects, which is neglected by most of the AMR models. It can be defined as the difference between maximum allowed flying time for each aircraft and the accumulated flying time since the last maintenance operation. Başdere and Bilge (2014) considered the remaining time in their developed model. The authors focused on the operational point of view that considered the stochasticity and the possibility of cancelling and delaying the flights. The problem was formulated as a multi-commodity network flow model, and it was solved by the use of both branch-and-bound and compressed annealing. The authors reported that compressed annealing outperformed branch-and-bound for large-scale problems, and it could find feasible solutions in minutes, which is important for the airline industry.

5.7 Other characteristics of AMR models

In this section, Table IV summarizes the AMR models' characteristics such as planning horizon, network representation, model formulation, solution procedure, and computational study. Also, maintenance checks and main considerations are presented in the last six columns of Table IV.

5.8 Discussion

After reviewing AMRP-related studies, we can see that from 1992 up to 2000, the researchers have paid much attention on proposing AMRP with tactical focus, as appeared in Kabbani and Patty (1992), Clarke *et al.* (1997), Gopalan and Talluri (1998), Talluri (1998), and Mak and Boland (2000). These studies were successful in determining unique rotations or sequence of flight legs to be repeated every day by each aircraft, but failed to consider some of the operational maintenance requirements, such as the restrictions on the flying time, and the restrictions on the takeoffs. Implementing these rotations practically is quite difficult due to lack of considering the operational requirements of the airline industry. Thus, the researchers move to propose AMRP with operational focus as appeared in the period from 2003 up to 2014. These studies aim to generate maintenance feasible routes to be flown by individual aircraft in the fleet, as shown in the studies by Sriram and Haghani (2003), Sarac *et al.* (2006), and Başdere and Bilge (2014). Sriram and Haghani (2003) proposed an operational model and effective solution method, but their solution method can handle only small-size problem with only 58 flights and 13 aircraft. This performance restricts their model applicability in reality, since in reality the number of flights might reach up to 3,000 flights. Similarly, in Sarac *et al.* (2006), they proposed a model based on the commonly used set-partitioning formulation. However, a drawback of this approach is that it generates an exponential number of routes which needs a sophisticated method to solve the model, and it fails to handle real and large-size problems. Lastly, in Başdere and Bilge (2014), the authors proposed an operational AMRP that shows good performance to handle large-scale problems. However, this study fails to consider one important operational requirements mandated by FAA, which is the maximum takeoffs between two successive maintenance operations.

In overall, we notice that AMRP received much attention from the scholars as show in Table IV. However, there are some research gaps that are discussed in the last section of this paper.

6. CSP

The CSP is considered the fourth category in our paper, and the last stage of the airline schedule planning process, and it assumes that all previous stages are solved in advance. CSP is one of the most studied problems compared to other airline planning problems for the following reasons: first, the crew cost is the second highest cost after the fuel cost, so optimizing the CSP can make a significant reduction on the total operational cost of airline companies. Second, there are many regulations and contractual issues that should be considered in this problem, which increases the model complexity. In order to avoid the high complexity in CSP, the problem is solved sequentially in two steps. The first step is called the crew pairing problem (CPP), which provides, at a minimal cost, a set of pairing that will cover exactly once all the flight legs in the schedule, while satisfying the regulations and contractual issues. The second step is called the crew rostering problem (CRP) that uses the solution of the first step as an input and provides the legal schedule for each crew member. To do that, there are many regulations and contractual restrictions that should be respected (Barnhart *et al.*, 2003). In the remainder of this section, we discuss the CPP and CRP and their related models in the literature, as shown in Sections 6.1 and 6.2, respectively. Also, in Section 6.3, we present models that integrate both problems partially or totally. Table V closes this section and presents the main features of the reviewed crew scheduling papers as shown in Section 6.4. In Section 6.5, we provide a discussion for reviewed CSP papers.

Table V. Characteristics of crew scheduling models

Author/s (year)	Problem Tackled		Planning Horizon	Network	Model formulation	Objective function	Solution procedure	Computational study		Other features				
	CPP	CRP						Data	Airline	BCPM	RM	SM	CS	CMS
Lavoie *et al.* (1988)	✓		NA	(DPN)	SC	Min pairing cost	Continuous relaxation and Column generation approach	RL	Air France	✓				
Anbil *et al.* (1991)	✓		D	(DPN)	SP	Min pairing cost	TRIP optimization system	RL	US					
Ryan (1992)		✓	M	–	Sp	Min rostering cost	ZIP optimization package	RL	Air New Zealand					✓
Graves *et al.* (1993)	✓		D	TSN (FN)	SP	Min pairing cost	Optimization system consists of generator and optimizer	RL	United Airlines					
Hoffman and Padberg (1993)	✓		NA	NA	SP	Min pairing cost	Branch-and-cut approach	RL	–	✓				
Barnhart *et al.* (1995)	✓		NA	(DPN)	SP	Min pairing and deadhead cost	Developed algorithm	RL	–	✓				
Chu *et al.* (1997)	✓		D	–	SP	Min pairing cost	Graph based branching heuristic	NA	–					
Day and Ryan (1997)		✓	W	–	SP	Min the rostering cost	ZIP optimization package	RL	Air New Zealand					✓
Desaulniers, Desrosiers, Dumas, Marc, Rioux, Solomon and Soumis (1997)	✓		W	TSN (FN)	MCNF	Min pairing cost	Branch-and-bound algorithm based on Dantzig-Wolfe decomposition principle	RL	Air France	✓				
Vance *et al.* (1997)	✓		D	(DPN)	DP	Min duty period and pairing costs	Linear programming relaxation and dynamic column generation	RL	–	✓				
Gamache *et al.* (1998)		✓	M	–	SP	Max the schedule score of crew member	Column generation embedded on a branch-and-bound-tree	RL	Air Canada				✓	

(continued)

Author/s (year)	Problem Tackled CPP	CRP	Planning Horizon	Network	Model formulation	Objective function	Solution procedure	Computational study Data	Airline	Other features BCPM	RM	SM	CS	CMS
Gamache *et al.* (1999)		✓	M	–	SP	Min cost of uncovered pairing	Column generation approach	RL	Air France					✓
Lučic and Teodorovic (1999)		✓	M	–	NA	Min average and absolute flight time deviation	Developed algorithm based on pilot-by-pilot heuristic algorithm and simulated annealing technique	RL&G	–				✓	
Dawid *et al.* (2001)		✓	M	–	SP	Max the total utility value	SWIFTROSTER algorithm	RL	European					
Klabjan *et al.* (2001)	✓		W	TSN (FN)	MCNF	Min pairing cost and regularity penalty	Developed algorithm	RL	United Airlines	✓				
Fahle *et al.* (2002)		✓	M	–	SP	Min rostering cost	Constraint programming based column generation approach	RL	European				✓	
Yan and Chang (2002)	✓		W	(DPN)	SP	Min the total crew cost	Column generation approach	RL	Taiwan	✓			✓	
Ehrgott and Ryan (2002)	✓		NA	–	SP	Min pairing cost and non-robustness	1st: Weighted sum method 2nd:ε-constraint method	RL	–		✓			
Cappanera and Gallo (2004)		✓	M	(DPN)	MCNF	Min number of uncovered activities	CPLEX 7.0 (ILOG 2000)	RL	Italian					✓
Schaefer *et al.* (2005)	✓		D	NA	SP	Min expected crew pairing cost	Push-back recovery heuristic	G	–			✓		

(*continued*)

Table V.

Table V.

Author/s (year)	Problem Tackled CPP	CRP	Planning Horizon	Network	Model formulation	Objective function	Solution procedure	Computational study Data	Airline	Other features BCPM	RM	SM	CS	CMS
Yen and Birge (2006)	✓		D	TSN (FN)	SP	Min pairing cost	Flight-pair branching algorithm	RL	Air New Zealand		✓	✓		
Guo *et al.* (2006)	✓	✓	NA	TSN (FN)	NF	Min overnight and compensation costs	Rostering heuristic	RL	European					
Shebalov and Klabjan (2006)	✓		D	TSN (FN)	SP	Max the number of move-up crews	Delayed column generation and Lagrangian relaxation	RL	–		✓			
Zeghal and Minoux (2006)	✓	✓	W&M	–	ILP	Min total number of extra flight credits.	1st: CPLEX 6.0.2 2nd: Developed heuristic based on a rounding strategy embedded in a partial tree search procedure	RL	Tunis Air				✓	
Achour *et al.* (2007)		✓	M	–	SP	Max the schedule score of crew member	Exact approach based on column generation	RL	North American carrier				✓	
Souai and Teghem (2009)	✓	✓	D	–	SP	Min total cost and deviation.	1st: Hybrid genetic algorithm 2nd: Two local search heuristic(Legality Repair Heuristic and Feasibility Repair Heuristic)	RL	Air-Algérie				✓	
Tekiner *et al.* (2009)	✓		NA	TSN (FN)	SP	Max number of desired pairing.	ILOG OPL Studio5.5/CPLEX 11.0	RL	Local Airline in Turkey		✓		✓	

(*continued*)

Author/s (year)	Problem Tackled		Planning Horizon	Network	Model formulation	Objective function	Solution procedure	Computational study		Other features				
	CPP	CRP						Data	Airline	BCPM	RM	SM	CS	CMS
(Maenhout and Vanhoucke, 2010)		✓	M	(DPN)	SP	Min total penalty cost	1st: Hybrid scatter search heuristic 2nd: Exact branch-and-price procedure 3rd: Steepest descent variable neighborhood search	RL	Brussels Airlines					✓
Deng and Lin (2011)	✓		D&W	TSN (FN)	TSP	Min pairing, rest, and under utility costs	Ant colony optimization algorithm	RL	–	✓			✓	
Saddoune *et al.* (2011)	✓	✓	M	TSN (FN)	SP	Min schedule and pilot penalty cost	Column generation and dynamic constraint aggregation method	RL	North American				✓	
Muter *et al.* (2013)	✓		D&W	TSN (FN)	SC	Min pairing and deadhead costs	Column generation approach	RL	Local Airline in Turkey		✓		✓	

Notes: Problem tackled: crew pairing problem (CPP) or crew rostering problem (CRP). Planning horizon: daily (D), weekly (W), or monthly (M). Network: duty period network (DPN) or time-space network (TSN) = flight network (FN). Model formulation: network flow or multi-commodity network flow (MCNF), set-partitioning (SP), set covering (SC), traveling salesman problem (TSP) or Duty period (DP). Used Data: real life (RL) or generated (G). Other features: basic crew pairing model (BCPM), robust model (RM), stochastic model (SM), cockpit scheduling (CS), cabin member scheduling (CMS)

Table V.

6.1 CPP

This is considered the first step of crew scheduling that must be solved in order to assign, at minimal cost, pairings to flight legs so that each flight in schedule is covered once, while considering the regulations posed by the FAA and the company rules, which may be different between airline companies. Each pairing or crew trip is usually a set of duties, starting and ending at the same crew base or hub. Each pairing should be assigned to crew members working in different positions, so that each flight is covered by the required attendants (pilot, copilot, stewards, and others). One restriction for assigning two consecutive flights to the same crew member is that the connection time between the two flights should be sufficient for the crew member to move from one flight to the other, but this constraint is relaxed if both flights are flown by the same aircraft. Actually, the CPP has received much attention since most of the cost benefit lies in this part of crew scheduling process. In the following sections, we categorize the CPP into subcategories based on the model characteristics.

6.1.1 Basic crew pairing models (CPM). Basic CPM are developed in order to assign the generated pairing to the scheduled flight legs, while neglecting the uncertainty issues that are discussed in the following sections. In this section, we review such models, which can be categorized into two main groups: the daily pairing problem and the weekly pairing problem. In the daily problems, it is assumed that the flight schedule is repeated every day of the week, while in the weekly problems, it is assumed that each day of the week has different flight schedule.

Starting with the weekly problems, Desaulniers, Desrosiers, Dumas, Marc, Rioux, Solomon, and Soumis (1997) presented a model that outperformed the system used by Air France due to considering accurate cost function. The proposed model considered the non-linear cost function without any approximation, which was neglected in other models, such as Lavoie *et al.* (1988), and Yan and Chang (2002). The paper by Klabjan *et al.* (2001) incorporated the idea of regularity in their model, meaning the crew will repeat itineraries, which eased the implementation and management processes. The model was solved by partitioning the legs into small groups and finding pairings in each group that can be repeated every day of the week. On the focus of cockpit pairing, Yan and Chang (2002) proposed a model for feasible pairing of the pilot, copilot, and flight engineer. For checking the applicability of their model, a case study was carried out in Taiwan Airways, regarding international operations covered by seven aircraft types serving 444 flights. The model provided solutions in less than ten seconds. Another weekly version of CPM appeared in the work by Deng and Lin (2011) who formulated the problem as a traveling salesman problem and solved it by using an ant colony optimization (ACO). ACO showed good potential tool to solve other scheduling problems in airline planning.

Moving to the daily problems, Hoffman and Padberg (1993) proposed Branch and Cut as a solution methodology for large-scale problems, which could not be solved by traditional approaches. On a trial to improve the performance of the CPM, Barnhart *et al.* (1995) proposed a model that improved crew utilization through using deadhead crews, especially for long-haul operation that have long inactivity time. The model showed a significant reduction in the annual crew cost, reaching $5 million. There were some research attempts aimed at improving the speed of solution and the overall performance of the CPM. For example, Vance *et al.* (1997) presented a new formulation that depends on decomposing the problem into two subproblems. The authors applied Dynamic Column Generation in order to reduce the number of variables and increase the speed of solution convergence.

6.1.2 CPM implemented by airlines. In the literature, there are some successful attempts that have been aimed to improve the existing airline systems. For example, Anbil *et al.* (1991) developed a SPIRIT approach to solve large-scale problems efficiently by the usage of local improvement heuristics. Using such a model in American Airlines, it realized annual savings of about $20 million. Also, Graves *et al.* (1993) developed a new system for United Airlines, which worked efficiently in solving medium- and large-sized problems. The company yielded $16 million as annual saving by using this system. In addition, Chu *et al.* (1997) developed a model that provided a good pairing with the adoption of branching heuristics. The success of this model motivated American Airlines to use it as a planning system, which yields savings of $2 million annually.

6.1.3 Stochastic CPM. In the previous two sections, all the reviewed models are deterministic models that neglect the uncertainty issue, which is the main focus of this section. One CPM that considers the uncertainty was presented by Schaefer *et al.* (2005) who developed a model that was better than the deterministic models. That model yielded good practical performance during the frictional disruptions over the deterministic models (e.g. the disruptions with limited duration such as weather changes and airport congestion). This work was further extended by Yen and Birge (2006) who proposed a two-stage stochastic model that captured proactively the interaction between long-range and short-range decisions. The computational results showed that airline companies can have significant savings when considering the uncertainty during the planning stage.

6.1.4 Robust CPM. The airline industry is often disrupted by external changes such as bad weather, technical problems, and others. So, developing robust pairing models that provide better solutions is really satisfying the needs of airline companies. In the paper proposed by Ehrgott and Ryan (2002), a robust model was developed for solving a model with two conflicting objectives, maximizing the robustness and minimizing the crew cost. They developed a bi-criteria optimization approach that provided a Pareto optimal solution. They represented the robustness by penalizing any connection that had limited time and may cause delay. The model was tested using real world data, and the results showed improvement in the robustness with a small increase in the cost. Another mechanism for robustness was presented by Shebalov and Klabjan (2006) who proposed a model that can provide robust solutions through maximizing the number of move-crews (crews that can be swapped). The model was solved by delayed column generation and Lagrangian relaxation. The model was evaluated by simulation; the model showed better performance over the traditional models in terms of robustness, cost, and number of flight cancellations. The model provided a schedule with many swapping opportunities.

Moreover, Tekiner *et al.* (2009) proposed the idea of extra flights as a source of disruption that should be managed at the planning stage. Swapping and deadhead options were used to manage these flights without making any delay or cancellation to the schedule. The authors proposed two robust integer programming models that provided robust solutions, but the major disadvantage of this model was that it solved only small-size problems. In order to solve large-scale problems, Muter *et al.* (2013) extended Tekiner *et al.*'s work to develop an efficient column generation approach that could tackle real problems. They developed two pruning rules to enhance the column generation approach that was used to solve the pricing subproblems.

6.2 CRP

The CRP is solved after the CPP in order to assign the generated pairing to each crew individually, taking into account vacations, crew requests, and skills, while considering the regulations, company rules, and union agreements. In this step, the crew schedule is generated so that all flights are covered by the required crew attendants (stewards and stewardess) and cockpit members (pilot, co-pilot, and flight engineer). The rosters are usually generated one month before the day of operation. The CRP is solved not only to minimize the operational cost but also to maximize the social quality for each crew member by considering their preferences. In order to construct the monthly crew schedule, there are two main methods: the rostering system and the preferential bidding system. In the following section, we discuss each method and the related work in the literature.

6.2.1 The rostering approach of CRP. The rostering approach is adopted by many European airlines such as Air France, Alitalia, Lufthansa, and Swiss Air. This system is also used by Air New Zealand. The rostering approach constructs monthly crew schedules by generating individual rosters, taking into account the pre-assignment activities and crew requests. This approach attempts to provide a crew schedule which has a fair and equal share among crew members. The rostering approach has received much attention compared to the other method.

Rostering approach appeared in some works that focused on cabin crew rostering. For example, Ryan (1992) proposed a model with the aim of providing equitable rosters. The efficiency of the proposed work was appreciated by Air New Zealand and was implemented in 1989. In addition, Day and Ryan (1997) proposed a model to avoid manual allocation of rosters in Air New Zealand during the assigning duties and days-off for each crew member. The model is decomposed into two parts: first, assigning off-days, and second assigning the pairings and other activities. Both submodels are solved by the adoption of column generation, but for each submodel, the total number of columns is quite limited compared to the case in which each submodel is optimized simultaneously. The model was tested using real data provided by Air New Zealand; the results indicated that this method led to good solution quality only for short-haul operations. Moreover, Gamache *et al.* (1999) extended Day and Ryan's work by proposing an accelerated Column Generation approach. The results showed significant reduction in the computational time by a factor of more than 1,000, with little deterioration in the solution quality.

On the other hand, rostering approach was used in papers with the focus on proposing meta-heuristics, specific algorithms, and new formulations. Regarding meta-heuristics, Lučic and Teodorovic (1999) used simulated annealing as a solution method for multi-objective problems. This method was tested by data instances provided by small and medium airline companies. With respect to specific algorithms, Dawid *et al.* (2001) developed a SWIFTROSTER algorithm to solve real world problems. This algorithm was successful in solving problem involving 1,300 crew members and six fleet types in reasonable computational time. On the focus of proposing new formulation different from set-partitioning formulation, Cappanera and Gallo (2004) formulated the rostering problem as a multi-commodity flow problem where the crew members represent the commodity. The model representation was tight, and the computational experiments showed good performance to handle a medium-size test instance delivered from Italian Airline.

More recently, Maenhout and Vanhoucke (2010) proposed a model that was quite different from the others. They presented a model that considered crew preference besides the pre-assignment activities, crew information, and regulations in order to maximize the social quality of the crew schedule. They used a hybrid scatter search approach in order to solve the problem that aimed to assign rosters to each crew member and decided if extra or freelance personnel should be hired.

While many research studies focused on cabin crew assignment as mentioned above, other research focused on pilot assignment by using the rostering approach. Fahle *et al.* (2002) presented a model that captured all complex regulations neglected by the other models. Each airline regulation was expressed in the model. The model considered a real life case during the computational experiments, comprising 400 crew members and 1,000 activities. This proposed model showed significant reduction in the computation time.

6.2.2 The preferential bidding approach of CRP. This approach is usually implemented by US and Canadian Airlines. Crew schedule can be constructed by this approach not only by considering the pre-assignment activities like the rostering approach but also the crew preference that is reflected by weighted bids. This approach has not received much attention from researchers, because most airline companies prefer to adopt the rostering system.

Gamache *et al.* (1998) presented a model for assigning pilots by using the preferential bidding method with the aim of maximizing an individuals' preference and the social quality of the schedule. The idea was to assign duties, days-off, and vacations by considering some weighted bids that reflected each individual's preference. The problem was solved for each crew member starting from the most senior and ending with the most junior. Each solution means selecting the schedule with the best score for each individual. Air Canada substituted their current system with the system generated by this research in 1995 because of its significant improvement over its old system. In a trial to improve Gamache *et al.*'s work, Achour *et al.* (2007) presented an exact method to solve the same problem. They improved the model by delaying the assignment of the best score schedule to the most senior member until there was only one remaining for that member. This method yielded a best score for the seniors without any effect of finding the best score for the others. This model was tested in real life problems and showed solution quality improvement over the existing methods.

6.3 Integrated CSP

The crew scheduling model is usually decomposed into two separate models, and each one is solved independently, leading to suboptimal solutions. Integration of the two models is a step forward to avoid suboptimal solutions and it improves the overall performance of the scheduling system. Recently, scholars have paid attention to solving integrated CSPs in two different ways, sequentially (partial integration) or simultaneously (full integration).

Solving integrated crew scheduling sequentially appeared in the work by Guo *et al.* (2006), who solved the pairing and rostering problems in a successive way. The model first selected the set of crew pairing, and then the model provided individual crew schedule by using the generated pairing. The authors reported that through this partial integration, there was a significant reduction in the total crew cost.

The sequential model has two main drawbacks, which are a lack of good cost estimation and neglecting the crew availability when a pairing problem is solved. These drawbacks motivated the researchers to solve the CSPs simultaneously as noticed by some scholars. For example, Zeghal and Minoux (2006) proposed a model for technical crew members (pilots and officers). In addition, Souai and Teghem (2009) proposed another model, while satisfying all the regulations and rules.

6.4 Other characteristics of crew scheduling models

Table V presents other characteristics of the reviewed models that were not discussed in the previous sections. Table V summarizes these characteristics, which are as follows: the planning horizon, type of network representation, objective function, solution procedure, and computational study. Moreover, other features of the models are presented in the last five columns of Table V.

6.5 Discussion

After discussing CSP paper, we observe that it is the most studied part in the airline schedule planning. CSP mainly contains two parts: crew pairing and crew rostering.

Starting with crew pairing, we noticed that it received much attention than crew rostering. The reason behind that most of cost reduction can be achieved by producing an efficient pairing that minimizes the operational cost. At the beginning, the researchers focus on studying CPP with daily planning horizon, such as Hoffman and Padberg (1993), Barnhart *et al.* (1995), and Vance *et al.* (1997). These studies show significant improvement to the solution methodology, since they can handle large-scale problems. However, that solution will not be completely feasible in practice, because it assumes that all flown flights are repeated every day of the week. In reality, building pairing based on this assumption in not applicable, because one or more flights do not fly on the weekend or particular day. Therefore, the researchers avoid this issue and develop pairing models based on weekly planning horizon, for example, Desaulniers, Desrosiers, Dumas, Marc, Rioux, Solomon and Soumis (1997), Yan and Chang (2002), and Klabjan *et al.* (2001). These studies avoid the limitations of the previous models and produce pairing to each specific day of the week. The discussed daily and weekly models assume that all departure times for the flights are fixed and know, which are proven to be wrong practically. Thus, the stochastic and robust crew pairing models start to appear to cover this point as shown by Ehrgott and Ryan (2002), Schaefer *et al.* (2005), Yen and Birge (2006), Shebalov and Klabjan (2006), Tekiner *et al.* (2009) and Muter *et al.* (2013). From practical side of view, the solution provided by these robust and stochastic models can be easily implemented in the real aviation industry.

Moving to the crew rostering, there are two approaches: the rostering approach and preferential bidding approach. Regarding rostering approach, Ryan (1992) presented a model that initially generates a set of feasible rosters and then solves the rostering problem. The performance of such approach depends on the quality of the rosters that are generated at the beginning. In another study, Day and Ryan (1997) used the standard column generation that fails to handle large-scale problems. The rostering approach was also investigated by other papers like Gamache *et al.* (1999), Lučic and Teodorovic (1999), Dawid *et al.* (2001), and Maenhout and Vanhoucke (2010). Although the rostering approach presented by the previous studies is implemented by many European airlines, it has one drawback. It does not give the individuals the chance to show their preference regarding the rosters resulting in assigning some rosters that are not preferred by those individual. On the other hand, the preferential bidding approach received less attention compared to rostering approach. It succeeds in avoiding the drawback of rostering approach, as shown by Gamache *et al.* (1998) and Achour *et al.* (2007).

We can see from Table V that the cabin member scheduling received less attention than cockpit scheduling. This is because the costs related to cabin crew are significantly lower than those of the cockpit crew and also because the problem related to cabin crew is a much large one.

7. Integrated airline schedule planning models

In the previous sections, each airline planning process is solved sequentially and independently of the other processes to avoid computational complexity. Actually, the main drawback of this way is that the optimal solution of one process is not optimal for the subsequent processes (suboptimal solution), which leads to infeasible inputs to the other processes. This main drawback is one of the main reasons that motivate researchers to focus on integrating more than one schedule planning process simultaneously, so as to ameliorate the solution quality and the anticipated profit of the airline companies. Although no attempts have been made to propose full integration of the four different planning processes, there are many integrated models of two or more different planning process that have been proposed. In this section, we present the last category that includes such integrated models, the solution techniques used for each model, and the implications of each model on the anticipated profit. In the last subsection, we provide discussion regarding the integrated models.

7.1 FSP integrated with FAP

Focused on the integration of FSP and FAP, Yan *et al.* (1997) proposed a model that adopted the option of aircraft rental in order to avoid demand fluctuation and market changes.

The proposed model was tested through a case study in a major Taiwan airline carrier, which covered 15 cities by 335 flights operated by 12 aircraft. The authors reported potential profit saving by using this model. Also, Ioachim *et al.* (1999) studied the weekly FAP and incorporated schedule synchronization constraints for long-haul operations. Their proposed model showed improvement in terms of the solution time, so it is efficient in tackling large-scale real problems.

The interaction issue between passenger demand and flight supply was studied by Yan and Tseng (2002) and Lohatepanont and Barnhart (2004). With respect to Yan and Tseng's work, the authors proposed a model that assigned the aircraft and passenger simultaneously with the objective of minimizing the total cost. They developed a solution methodology that helped the model to converge within a 3 percent error gap. On the other hand, Lohatepanont and Barnhart (2004) presented a model that considered many aspects that were neglected by Yan and Tseng's model. These aspects are, for example, considering different fare class instead of aggregating it into one fare class, demand recapture, and flight leg interdependencies.

In an attempt to present a tight representation for FSP and FAM, Sherali *et al.* (2009) performed polyhedral analysis and bender decomposition for their proposed model. They considered itinerary-based demand and different fare classes. The authors validated their proposed model by using the data from US Airline, and using such a model resulted in $28.3 million increase in annual profit.

7.2 FAP integrated with AMRP

This kind of integration takes into account the interdependency between the two problems which leads to better and near-optimal solutions. For the joint aircraft fleeting and routing problem, Haouari *et al.* (2009) proposed a network flow-based heuristics to solve their proposed integrated model. On the set of real data acquired from Tunisair, the model could find very near-optimal solutions in less than eight seconds.

Another FAP integrated with AMRP was reported by Zeghal *et al.* (2011) who incorporated the idea of renting out some aircraft during low-demand seasons while hiring some aircraft during high-demand periods. The model considered the fleet heterogeneity and long-term maintenance issues. The authors reported that the model realized a $33.8 million increase in annual profit when testing the model by using real life data from Tunisair. Short-term maintenance and long-term maintenance integrated with FAP were considered by Haouari *et al.* (2011) who presented an exact solution approach to solve the integrated model, which was solved by using branch-and-price approach and benders decomposition. The first approach showed better performance over the second in finding the optimal solution, while the second was better in finding fast and near-optimal solutions. The model could find optimal solution to a problem instance comprised of 507 rotations, 1,050 flight legs, and 34 aircraft from different types. However, the test instances are not big scale enough to test the applicability of the proposed methods in real industry. Weekly FAP integrated with AMRP was studied by Liang and Chaovalitwongse (2012) who proposed an innovative rotation tour network for modeling the problem. Recently, Dong *et al.* (2016) presented a heuristic approach to solve the integrated model that considered the price elasticity of each itinerary. The results showed an achievement of significant profit improvement in a reasonable computation time.

7.3 AMRP integrated with CPP

The integrated AMRP and CPP models aim to determine, at minimum cost, a set of flight legs that are covered exactly by one aircraft and one crew base, while satisfying the maintenance requirements and other side constraints. Based on the AMRP integrated with CPP, Cordeau *et al.* (2001) proposed a model that considered the maintenance issues and minimum connection time. To handle this kind of formulation, bender decomposition and column generation were used as the solution method. In a test instance comprising 500 legs, the proposed model showed better crew cost reduction (9.4 percent) over the sequential approach. Mercier *et al.* (2005) improved the model stated by Cordeau *et al.* to provide some penalties for connections that may delay, which in turn led to the improvement of the solution robustness. The authors used two different bender decomposition methods, and the Pareto cut method was studied to improve the speed of solution convergence.

Providing robust AMRP integrated with CPP was presented by Weide *et al.* (2010). The authors tried to avoid the short and restricted connects and used them as a mechanism for robustness. To solve such model, the authors used an iterative approach that stopped the iterations when there was no more opportunity to improve the solution robustness. Based on real world data, the model provided solutions with low cost and high robustness. Another robust model was discussed by Dunbar *et al.* (2014) who incorporated stochastic delay information to their integrated model. The authors improved the robustness by minimizing the propagated delay. They proposed two algorithms to reduce the delay propagation by re-timing the routing and crew scheduling. The practical viability of the model was tested, and the computational results showed about 14 percent as an average reduction in the delay propagation.

All the previous papers in this section can be considered as a full integration between AMRP and CPP, which means the objective function has decision variables related to each problem. The next two papers attempted to provide partial integration of AMRP and CPP. The first attempt was introduced by Klabjan *et al.* (2002) who incorporated plane count constraints so as to provide a crew schedule which maintained the maintenance feasibility at the same time. The model provided significantly better solutions than traditional pairing models. Another attempt for partial integration emerged in a study by Cohn and Barnhart (2003). The authors incorporated the maintenance issues in order to improve the performance of the crew scheduling model and reduce the total cost. This model guaranteed the maintenance feasible crew pairing solution.

7.4 FAP integrated with CPP

In the previous sections, scholars tried to integrate two consecutive planning processes, but in this section, we discuss research works that integrate two non-consecutive planning problems such as FAP and CPP. For example, Sandhu and Klabjan (2007) proposed a full integration of fleeting and crew pairing, while neglecting the maintenance issues. The plane count constraint was used in order to capture the routing issue. The proposed model was solved in two different ways: first, by the use of Lagrangian relaxation coupled with column generation, while the benders decomposition was acquired in the second method. The results showed a reduction of about 3 percent in the overall cost, and the performance of Lagrangian relaxation approach outperformed the benders decomposition method. Another model integrating FAP with CPP appeared was presented in a work by Gao *et al.* (2009), who incorporated the station purity to their model by limiting both fleet types and available crew base in any airport. The proposed model was validated, and the results showed potential crew cost reduction ($4-8 million) besides amelioration of the solution robustness.

7.5 Integration of three different planning problems

In this section, we discuss research that aims to integrate more than two different airline planning processes. The complexity of these models is much higher than the others, but it provides better quality and near-optimal solutions if compared with the previous models. The main challenge related to these models is how to find an appropriate solution approach in order to tackle the complexity of such models and to find the solution in a reasonable computational time, which is essential for the airline industry.

In a trial that integrated FSP, AMRP, and CPP, Mercier and Soumis (2007) introduced a model that permits some flexibility in the departure time. The model was solved by the application of the benders decomposition method coupled with dynamic constrained generation. The computational results showed reduction of crew cost and reduced aircraft usage.

Integrating the first three planning processes, FSP, FAP, and AMRP, received attention from Sherali *et al.* (2013b). They proposed a complex model that considered many issues such as re-timing, demand recapture, maintenance issues, the through flights, and multiple fare classes. The high complexity of the model was handled by the application of the benders decomposition approach, and the computational results showed a 3 percent increase in the annual profit when compared with the traditional models. Also, Gürkan *et al.* (2016) proposed a model that considered the cruise speed for the first time in order to increase the fuel utilization and decrease the number of needed aircraft.

On the other hand, there are some scholars interested in integrating FAP, AMRP, and CPP. For example, Salazar-González (2014) proposed a model that was implemented by a funded project in Spain to operate daily flights. The authors presented a new compact formulation for the model that was quite similar to the two depot vehicle routing problem, and it provided an automatic approach to solve the problem instead of the efforts exerted by the planning mangers. Another attempt appeared by Shao *et al.* (2017) who proposed a model with itinerary demands considerations and developed some novel accelerating strategies for the benders decomposition approach in order to solve the proposed model. Recently, Cacchiani and Salazar-González (2016) developed a novel model, where arc-based variables represent the aircraft routes and path-based variables represent crew pairings. The developed model was solved using two methods, called path-path and arc-path methods. The results showed that the second method outperformed the first one in finding the optimal solutions, while solving a real test case up to 172 flights.

7.6 Other features of the reviewed integrated models

In this section, Table VI summarizes all the features of the reviewed integrated models, which are as follows: the planning horizon, type of network representation, objective function, the solution procedures, and the computational study.

7.7 Discussion

In the last two decades, we notice that the researchers have been interested in the integration of different airline problems. It is observed that the integration between AMRP and other phases received much attention from the scholars, while AMRP was discussed from the tactical point of view (Cordeau *et al.*, 2001; Haouari *et al.*, 2009; Weide *et al.*, 2010; Zeghal *et al.*, 2011; Haouari *et al.*, 2011; Liang and Chaovalitwongse, 2012; Dunbar *et al.*, 2014: Dong *et al.*, 2016). These studies succeeded in avoiding the suboptimality problem, but they fail to generate routes that are applicable in reality due to the lack of operational considerations.

Recently, it has been noticed that cruise speed in one of the mechanisms used to produce robust models (Gürkan *et al.*, 2016). For this approach, the cruise speed can be adjusted in order to decrease the flying time and reduce the delay propagation. Although there is efficacy of this approach theoretically, but practically, it increases the fuel cost as the fuel consumption is increased.

With the focus on robust AMRP integrated with CPP, Weide *et al.* (2010) and Dunbar *et al.* (2014) presented a model where AMRP was formulated based on the expected value of the non-propagated delay. Since the non-propagated delay is characterized by high level of uncertainty, these expected values may not properly reflect the final realization of the non-propagated delay, when implemented in reality. This will, in turn, lead to easily delay propagation, disrupted routes, and nonrobust solution.

Table VI. Characteristics of integrated schedule planning models

Author/s (year)	Problem Tackled				Planning horizon	Network	Model formulation	Objective function	Solution procedure	Computational study	
	FSP	FAP	AMRP	CPP						Data	Airline
Clarke *et al.* (1996)		✓	✓	✓	D	TSN	ILP	Min the assignment cost	Developed algorithm	RL	US
Yan *et al.* (1997)	✓	✓			W	TSN	NF	Min fleeting and flight scheduling cost	Lagrangian based algorithm, subgradient method, the network simplex method and a Lagrangian heuristic	RL	Taiwan
Barnhart *et al.* (1998)		✓	✓		W	CN	SP	Min maintenance cost of selected strings	Branch and price, and cut algorithm	G	–
Ioachim *et al.* (1999)	✓	✓			W	CN	MCNF	Min the assignment cost	Linear relaxation by Dantzig-Wolfe decomposition and column generation embedded within a branch-and-bound method	RL	European
Cordeau *et al.* (2001)			✓	✓	D	TSN	MILP	Min routing and crew cost	Benders decomposition, column generation, and branch-and-bound heuristic	RL	Canadian
Yan and Tseng (2002)	✓	✓			NA	TSN	MCNF	Min the fleeting and passenger cost	algorithm based on Lagrangian relaxation, a sub-gradient method, the network simplex method, the least cost flow augmenting algorithm and the flow decomposition algorithm	RL	Taiwan
Klabjan *et al.* (2002)			✓	✓	D	TSN	SP	Min pairing cost	Developed algorithm	RL	United Airlines
Cohn and Barnhart (2003)			✓	✓	NA	TSN	SP	Min pairing cost	Branch and price approach	NA	–
Lohatepanont and Barnhart (2004)	✓	✓			D	NA	MILP	Max revenue minus operating and spill cost plus recaptured revenue	Developed algorithm based on column and row generation, and branch-and-bound	RL	US
Mercier *et al.* (2005)			✓	✓	D	TSN	MILP	Min crew, routing and penalty costs	Benders decomposition methods	RL	–

(*continued*)

Author/s (year)	Problem Tackled FSP	FAP	AMRP	CPP	Planning horizon	Network	Model formulation	Objective function	Solution procedure	Computational study Data	Airline
Mercier and Soumis (2007)	✓		✓	✓	D	NA	MILP	Min routing and crew cost	Benders decomposition method with a dynamic constraint generation	RL	–
Sandhu and Klabjan (2007)		✓		✓	D	TSN	MILP	Max fleet revenue minus crew cost	1st: Combination of Lagrangian relaxation and column generation 2nd : Benders decomposition approach	RL	US
Gao *et al.* (2009)		✓		✓	NA	TSN	MILP	Max fleet profit minus crew and penalty cost	ILOG CPLEX 9.0	RL	US
Haouari *et al.* (2009)		✓	✓		W	CN	MCNF	Min assigning and routing cost	Network flow-based heuristic	RL	Tunis Air
Sherali *et al.* (2009)	✓	✓			NA	NA	MILP	Max the revenue	Benders' decomposition method	RL	US
Weide *et al.* (2010)			✓	✓	W	TSN	SP	Min routing, crew, and crew penalty cost	Developed Algorithm	RL	–
Zeghal *et al.* (2011)		✓	✓		W	TSN	SP	Max net profit	Tailored optimization-based heuristics	RL	Tunis Air
Haouari *et al.* (2011)		✓	✓		W	TSN	SP	Min the assigning and deadhead cost.	1st: Benders decomposition 2nd: Branch-and-price	RL	Tunis Air
Liang and Chaovalitwongse (2012)		✓	✓		W	TSN	NF	Max the profit	1st: Diving heuristic 2nd: CPLEX callable library version 12.1	RL	US
Pita *et al.* (2012)	✓	✓			D	NA	MILP	Max the profit	Xpress with the optimizer version 20.00.11 (FICO 2009)	RL	TAP Portugal
Sherali *et al.* (2013a, b)	✓		✓		D	CN	MILP	Max the net profit	Reformulation-linearization technique and benders' decomposition-based method	RL	United Airline
Cadarso and Marín (2013)	✓	✓			D	NA	MCNF	Max the profit	–	RL	Spanish
Dunbar *et al.* (2014)			✓	✓	D	CN	MILP	Min total delay propagation cost	Re-timing heuristic	RL	–

(*continued*)

Table VI.

Table VI.

Author/s (year)	Problem Tackled FSP	FAP	AMRP	CPP	Planning horizon	Network	Model formulation	Objective function	Solution procedure	Computational study Data	Airline
Díaz-Ramírez *et al.* (2014)			✓	✓	D	CN	SP	Min routing, pairing and crew deadhead cost	Developed Algorithm	RL	Latin American
Salazar-González (2014)		✓	✓	✓	D	CN	MILP	Min fleeting, and routing and crew costs	Developed Algorithm	RL	Canary Island
Shao *et al.* (2017)		✓	✓	✓	D	CN	NLMIP	Max overall profit	Accelerated benders decomposition approach	RL	US
Gürkan *et al.* (2016)	✓	✓	✓		D	CN	NLMIP	Min spill cost, fuel consumption, and CO_2 emissions	1st: Discretized approximation and cruise speed control algorithm 2nd: Multi-stage triplet search algorithm	RL	US
Dong *et al.* (2016)	✓	✓			D	TSN	MCNF	Min aircraft operating cost	Relaxation heuristic algorithm	RL	Chinese carrier
Cacchiani and Salazar-González (2016)		✓	✓	✓	W	CN	MILP	Min weighted sum of number of aircraft routes, number of crew pairing, and waiting time of crews between flights	1st: Path-path method 2nd: Arc-path method	RL	Canary Island

Notes: Problem tackled: flight scheduling problem (FSP), fleet assignment problem (FAP), aircraft maintenance routing problem (AMRP), or crew pairing problem (CPP). Planning horizon: daily (D) or weekly (W). Network: connection network (CN) or time-space network (TSN). Model formulation: network flow (NF), multi-commodity network flow (MCNF), set-partitioning (SP), mixed integer linear programming (MILP) or non-linear mixed integer programming (NLMIP). Used Data: real life (RL)

8. Conclusion and future research directions

This paper aims to present an up-to-date review for airline schedule planning process. We categorize and summarize the trends and development of the relevant papers in order to support future research directions. Accordingly, we conduct our survey by using the following keywords "Flight scheduling," "Fleet assignment," "AMR," and "Crew scheduling." After screening and rejecting unelected research works, 116 research works were left that satisfied our scope for this paper.

First of all, we have categorized the airline schedule planning into four main categories, which are flight scheduling, fleet assignment, aircraft maintenance routing, crew scheduling, and integrated models. Then, based on the problem characteristics and operational considerations, a total of two sub-categories in flight scheduling, six sub-categories in fleet assignment, six subcategories in AMR, seven sub-categories in crew scheduling, and five sub-categories in integrated models are further identified. From the analysis, we notice that the most studied category is crew scheduling, followed by fleet assignment, then AMR, and finally the flight scheduling. Recently, in the period from 2000 up to now, the researchers have paid much attention toward studying the integrated models as they are more realistic and critical, especially from the aviation's perspective.

Finally, after discussing the research work done in airline schedule planning, several work avenues can be identified as future research directions as follows:

(1) For FSP:

- Re-timing the departure time has been demonstrated to be a promising approach for robust flight scheduling (Lee *et al.*, 2007). Here, the only drawback is assuming that rotation and crew issues are fixed and unchanged, since using such approach, the model will produce robust flight schedule, but for crew and rotations, the solution will be nonrobust. This point limits the applicability of that model in real aspect. Therefore, we suggest constructing a robust model that considers the variability of rotations and crew and solves them simultaneously. So, the generated solution will be robust. Consequently, the reliability of the flight scheduling can be increased.
- The market competition is considered in the flight scheduling by incorporating passenger choice behavior, which reflects the company's market share (Yan *et al.*, 2007). The market scope of that study is quite limited, because in the market, each company keeps changing their strategy after the knowledge of the others' market share, until reaching a stability point where each company maximizes its profit. In order to analyze opponents' actions in the market, we suggest using the dynamic Stackelberg game strategy for broad consideration of the market state. This would result in higher profitability for airline companies.
- Dynamic flight scheduling is one of approaches used to handle the stochastic passenger demand (Jiang and Barnhart, 2009). As discussed in that paper, the original timetable is adjusted to match the stochastic demand. Here, this adjustment might create some conflicts between flight scheduling and subsequent phases such as maintenance and crew plans, that were built based on the original timetable. Therefore, we propose evaluating the impact of dynamic scheduling to the maintenance and crew plans, in order to avoid that conflict. Consequently, the generated timetable will be more reliable and stable.

(2) For FAP:

- Re-fleeting is an approach that is used by airlines in response to demand uncertainty in order to change the initial fleet assignment based on updated demand information. Most of re-fleeting models are limited to homogenous fleet due to crew concerns (Berge and Hopperstad, 1993; Sherali *et al.*, 2005). The only drawback is that the re-fleeting possibilities are limited because all the aircrafts belong to the same family, resulting in inflexible and inefficient re-fleeting process. On the other hand, if, for instance, the re-fleeting is discussed while the fleet is heterogeneous, the number of re-fleeting possibilities will be increased, which in turn increases the process flexibility, but the crew consideration scope will be broaden. Accordingly, in the future research, discussing re-fleeting process under heterogeneous fleet is promising as it increases the efficiency of the re-fleeting process.

(3) For AMRP:

- The operational AMRP aims to generate maintenance feasible route. A route is maintenance feasible if it satisfies operational maintenance requirements such as the maximum number of days between successive maintenance operations, restrictions on the total accumulated flying hour, the total number of takeoffs, and the workforce capacity of the maintenance stations. To our knowledge, there are only two studies that have tried to consider most of the requirement (Barnhart *et al.*, 1998; Haouari *et al.*, 2012), but these studies fail to consider workforce capacity. In this situation, the model might schedule aircraft to receive maintenance in one station with insufficient capacity, resulting in prolonging the maintenance time, leading to a delay in covering the next flights. Therefore, proposing the model to consider all these requirements in one model will be appreciated by the airline companies, since the model will be more applicable.
- Currently, all AMRP models are discussed based on deterministic arrival and departure times, as shown by Sriram and Haghani (2003), Sarac *et al.* (2006), and Başdere and Bilge (2014). Practically, this assumption is proven to be wrong, since airline industry is often disrupted by external changes, causing changes in the arrival and departure times. Accordingly, in future research, proposing an AMRP model with stochastic time consideration is a promising direction as the robustness of the generated routes will be enhanced. To our best knowledge, it will be the first stochastic AMRP.
- The maintenance operation for aircraft is frequent and inevitable due to the regulation mandated by FAA. If, for instance, the model considers the capacity of the maintenance station as a fixed parameter (Sriram and Haghani, 2003; Sarac *et al.*, 2006), and for any reason there are shortages in that capacity, the aircraft will fail to resume its duty after finishing the maintenance operation, resulting in higher operational cost to find another aircraft to substitute the original one. Thus, the impact of delay in the maintenance station becomes very significant. Therefore, the uncertainties of the maintenance capacity should also be investigated in the future.

(4) For CSP:

- The rostering problem aims to assign each crew to the previously generated pairing in order to construct the rosters, while the number of crew members is fixed (Gamache *et al.*, 1999; Lučic and Teodorovic, 1999; Dawid *et al.*, 2001; Maenhout and Vanhoucke, 2010). The airline industry is most likely to face disruptions and unforeseen circumstances, as it might happen that the number of crew members decrease due to absence or illness of some members. In this situation, the whole roster will be affected and the airline should find another member to cover the pairing, resulting in higher operational cost and disruption to the original roster. Thus, the effect of uncertainties of the crew members during the rostering is obvious and critical. Therefore, considering this point is a fruitful research direction.

- The integrated crew scheduling models are usually solved by adoption of the rostering approach for crew assignment (Zeghal and Minoux, 2006; Souai and Teghem, 2009). Although the rostering approach presented is implemented by many European airlines, it has one drawback. It does not give the individuals the chance to show their preference regarding the rosters, resulting in assigning some rosters that are not preferred for those individual. For a better social quality schedule, we suggest developing an integrated scheduling model, while considering the preferential bidding system for building the rosters, which provides each crew with the opportunity to state their preference in regard to the assignment.

(5) For integrated airline schedule planning problems:

- The integrated AMRP with other phases was discussed based on tactical point of view for AMRP (Cordeau *et al.*, 2001; Haouari *et al.*, 2009; Weide *et al.*, 2010; Zeghal *et al.*, 2011; Haouari *et al.*, 2011; Liang and Chaovalitwongse, 2012; Dunbar *et al.*, 2014; Dong *et al.*, 2016). These studies succeeded in avoiding the suboptimality problem, but they failed in generating routes that are applicable in reality due to the lack of the operational considerations. Therefore, we suggest proposing integration while AMRP discussed from its operational point of view. It will result in generating maintenance feasible routes that can be easily implemented in real life aspect.
- Although all of the integrated schedule planning models have been studied, the challenge remains in studying the full integration of all four phases of airline schedule planning. In taking into consideration this connection, the proposed model will be more realistic and critical, but the complexity of the model will be increased. Therefore, the necessity of a sophisticated solution methodology is important in this situation.

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